One Hundred ACT, CLARIFY and REFUSE Examples

Experiment index

Question and Context

Could a larger unique BAYMAX contribution improve on the fixed 150-example pool without oversampling exact duplicates?

Change

The 1,000-row mixture targeted 300 unique BAYMAX examples: 100 productive ACT, 100 CLARIFY and 100 REFUSE, plus 700 generic xLAM productive examples.

"Balanced BAYMAX" is not "balanced full dataset": the xLAM portion still adds ACT targets, including multi-tool sequences.

Results

Evaluation	Cases	Task success	Tool accuracy	Argument accuracy	Clarification wording	Refusal wording
100/100/100 BAYMAX	150	43/150 (0.287)	0.567	0.761	0.050	0.172
100/100/100 BAYMAX	30	6/30 (0.200)	0.600	0.800	0.200	0.150
100/100/100 BAYMAX / strict replay	150	42/150 (0.280)	0.500	0.694	0.013	0.172
100/100/100 BAYMAX / strict replay	30	6/30 (0.200)	0.533	0.733	0.200	0.100

Values are ratios from 0 to 1, rounded to three decimals. Clarification/refusal columns measure wording criteria, not routing alone. Different suite sizes remain separate evaluations.

Routing and Output Validity

Evaluation	Macro-F1	Invalid-output rate
100/100/100 BAYMAX / strict replay (150 cases)	0.367	0.113
100/100/100 BAYMAX / strict replay (30 cases)	0.435	0.100

Routing values use the existing comparison helper on recorded per-scenario decisions; missing historical aggregate fields are derived without rewriting source files. Invalid output and unknown decision are not identical categories.

Result Provenance

- 100/100/100 BAYMAX: 2026-09-27 UTC, 150 cases.
- 100/100/100 BAYMAX: 2026-09-27 UTC, 30 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 150 cases.
- 100/100/100 BAYMAX / strict replay: 2026-09-28 UTC, 30 cases.

Interpretation

The historical full evaluation recorded 43/150 successes and the shift suite 6/30. Strict-parser replay reduced the full count to 42/150 and some argument/wording metrics.

This was a useful improvement over some early mixes, but the shift suite still exposed poor generalization and output-contract errors.

Limits

The historical and strict-replay rows are not different trained models. Use strict replay when comparing with the other replayed candidates. Old synthetic split and LoRA seed issues were not fixed by rescoring.

Supporting Files

The result links above are the surviving primary evidence. A separate immutable training record was not established for this historical experiment.

Notes assembled on 2026-10-10 from saved artifacts and repository records. No training, inference or result modification was performed to write this note.

PDF exported 2026-10-12. Original Markdown and benchmark evidence are preserved. Local artifact links require the project files.